SG No. 192 D^2_{6h} $P6/mcc$ [hexagonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: 2a, site symmetry: 622

Table 1: Wyckoff bond: 2a@2a, bond symmetry: 622

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, \frac{1}{4}]$	[1,2,3,4,5,6,-7,-8,-9,-10,-11,-12]
2	$[0, 0, -Z]$	$[0, 0, \frac{3}{4}]$	[13,14,15,16,17,18,-19,-20,-21,-22,-23,-24]

Table 2: Wyckoff bond: 6b@2a, bond symmetry: 222

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[0, 0, \frac{1}{4}]$	[1,-4,-8,11]
2	$[-2X, -X, 0]$	$[0, 0, \frac{1}{4}]$	[2,-5,-7,10]
3	$[X, -X, 0]$	$[0, 0, \frac{1}{4}]$	[3,-6,-9,12]
4	$[-X, -2X, 0]$	$[0, 0, \frac{3}{4}]$	[13,-16,-20,23]
5	$[2X, X, 0]$	$[0, 0, \frac{3}{4}]$	[14,-17,-19,22]
6	$[-X, X, 0]$	$[0, 0, \frac{3}{4}]$	[15,-18,-21,24]

Table 3: Wyckoff bond: 6c@2a, bond symmetry: 222

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, \frac{1}{4}]$	[1,-4,8,-11]
2	$[0, X, 0]$	$[0, 0, \frac{1}{4}]$	[2,-5,7,-10]
3	$[-X, -X, 0]$	$[0, 0, \frac{1}{4}]$	[3,-6,9,-12]
4	$[-X, 0, 0]$	$[0, 0, \frac{3}{4}]$	[13,-16,20,-23]
5	$[0, -X, 0]$	$[0, 0, \frac{3}{4}]$	[14,-17,19,-22]
6	$[X, X, 0]$	$[0, 0, \frac{3}{4}]$	[15,-18,21,-24]

Table 4: Wyckoff bond: 12d@2a, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, \frac{1}{4}]$	[1,-4]
2	$[-Y, X-Y, 0]$	$[0, 0, \frac{1}{4}]$	[2,-5]
3	$[-X+Y, -X, 0]$	$[0, 0, \frac{1}{4}]$	[3,-6]
4	$[Y, X, 0]$	$[0, 0, \frac{1}{4}]$	[7,-10]
5	$[X-Y, -Y, 0]$	$[0, 0, \frac{1}{4}]$	[8,-11]
6	$[-X, -X+Y, 0]$	$[0, 0, \frac{1}{4}]$	[9,-12]
7	$[-X, -Y, 0]$	$[0, 0, \frac{3}{4}]$	[13,-16]
8	$[Y, -X+Y, 0]$	$[0, 0, \frac{3}{4}]$	[14,-17]
9	$[X-Y, X, 0]$	$[0, 0, \frac{3}{4}]$	[15,-18]
10	$[-Y, -X, 0]$	$[0, 0, \frac{3}{4}]$	[19,-22]
11	$[-X+Y, Y, 0]$	$[0, 0, \frac{3}{4}]$	[20,-23]
12	$[X, X-Y, 0]$	$[0, 0, \frac{3}{4}]$	[21,-24]

Table 5: Wyckoff bond: `12e@2a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[0, 0, \frac{1}{4}]$	`[1,-8]`
2	$[-2X, -X, Z]$	$[0, 0, \frac{1}{4}]$	`[2,-7]`
3	$[X, -X, Z]$	$[0, 0, \frac{1}{4}]$	`[3,-9]`
4	$[-X, -2X, Z]$	$[0, 0, \frac{1}{4}]$	`[4,-11]`
5	$[2X, X, Z]$	$[0, 0, \frac{1}{4}]$	`[5,-10]`
6	$[-X, X, Z]$	$[0, 0, \frac{1}{4}]$	`[6,-12]`
7	$[-X, -2X, -Z]$	$[0, 0, \frac{3}{4}]$	`[13,-20]`
8	$[2X, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[14,-19]`
9	$[-X, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[15,-21]`
10	$[X, 2X, -Z]$	$[0, 0, \frac{3}{4}]$	`[16,-23]`
11	$[-2X, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[17,-22]`
12	$[X, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[18,-24]`

Table 6: Wyckoff bond: `12f@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, \frac{1}{4}]$	`[1,-11]`
2	$[0, X, Z]$	$[0, 0, \frac{1}{4}]$	`[2,-10]`
3	$[-X, -X, Z]$	$[0, 0, \frac{1}{4}]$	`[3,-12]`
4	$[-X, 0, Z]$	$[0, 0, \frac{1}{4}]$	`[4,-8]`
5	$[0, -X, Z]$	$[0, 0, \frac{1}{4}]$	`[5,-7]`
6	$[X, X, Z]$	$[0, 0, \frac{1}{4}]$	`[6,-9]`
7	$[-X, 0, -Z]$	$[0, 0, \frac{3}{4}]$	`[13,-23]`
8	$[0, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[14,-22]`
9	$[X, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[15,-24]`
10	$[X, 0, -Z]$	$[0, 0, \frac{3}{4}]$	`[16,-20]`
11	$[0, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[17,-19]`
12	$[-X, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[18,-21]`

Table 7: Wyckoff bond: `24g@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, \frac{1}{4}]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, \frac{1}{4}]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, \frac{1}{4}]$	`[3]`
4	$[-X, -Y, Z]$	$[0, 0, \frac{1}{4}]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, 0, \frac{1}{4}]$	`[5]`
6	$[X-Y, X, Z]$	$[0, 0, \frac{1}{4}]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{4}]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[0, 0, \frac{1}{4}]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[0, 0, \frac{1}{4}]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{4}]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, \frac{1}{4}]$	`[11]`
12	$[X, X-Y, -Z]$	$[0, 0, \frac{1}{4}]$	`[12]`

continued ...

Table 7

No.	vector	center	mapping
13	$[-X, -Y, -Z]$	$[0, 0, \frac{3}{4}]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[0, 0, \frac{3}{4}]$	`[14]`
15	$[X-Y, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[15]`
16	$[X, Y, -Z]$	$[0, 0, \frac{3}{4}]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[0, 0, \frac{3}{4}]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[18]`
19	$[-Y, -X, Z]$	$[0, 0, \frac{3}{4}]$	`[19]`
20	$[-X+Y, Y, Z]$	$[0, 0, \frac{3}{4}]$	`[20]`
21	$[X, X-Y, Z]$	$[0, 0, \frac{3}{4}]$	`[21]`
22	$[Y, X, Z]$	$[0, 0, \frac{3}{4}]$	`[22]`
23	$[X-Y, -Y, Z]$	$[0, 0, \frac{3}{4}]$	`[23]`
24	$[-X, -X+Y, Z]$	$[0, 0, \frac{3}{4}]$	`[24]`

* Wyckoff site: `2b`, site symmetry: `6/m..`

Table 8: Wyckoff bond: `2a@2b`, bond symmetry: `6/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,3,4,5,6,-13,-14,-15,-16,-17,-18]`
2	$[0, 0, -Z]$	$[0, 0, \frac{1}{2}]$	`[7,8,9,10,11,12,-19,-20,-21,-22,-23,-24]`

Table 9: Wyckoff bond: `6b@2b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-4,-13,16]`
2	$[-Y, X-Y, 0]$	$[0, 0, 0]$	`[2,-5,-14,17]`
3	$[-X+Y, -X, 0]$	$[0, 0, 0]$	`[3,-6,-15,18]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{2}]$	`[7,-10,-19,22]`
5	$[X-Y, -Y, 0]$	$[0, 0, \frac{1}{2}]$	`[8,-11,-20,23]`
6	$[-X, -X+Y, 0]$	$[0, 0, \frac{1}{2}]$	`[9,-12,-21,24]`

Table 10: Wyckoff bond: `12c@2b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-13]`
2	$[-Y, X-Y, Z]$	$[0, 0, 0]$	`[2,-14]`
3	$[-X+Y, -X, Z]$	$[0, 0, 0]$	`[3,-15]`
4	$[-X, -Y, Z]$	$[0, 0, 0]$	`[4,-16]`
5	$[Y, -X+Y, Z]$	$[0, 0, 0]$	`[5,-17]`
6	$[X-Y, X, Z]$	$[0, 0, 0]$	`[6,-18]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{2}]$	`[7,-19]`
8	$[X-Y, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[8,-20]`
9	$[-X, -X+Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[9,-21]`
10	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{2}]$	`[10,-22]`

continued ...

Table 10

No.	vector	center	mapping
11	$[-X+Y,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[11,-23]`
12	$[X,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[12,-24]`

* Wyckoff site: `4c`, site symmetry: `3.2`

Table 11: Wyckoff bond: `4a@4c`, bond symmetry: `3.2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4,5,6,-7,-8,-9]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[13,14,15,-22,-23,-24]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[16,17,18,-19,-20,-21]`

Table 12: Wyckoff bond: `12b@4c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,-11]`
2	$[0,\ X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[2,-10]`
3	$[-X,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[3,-12]`
4	$[-X,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[5,-7]`
6	$[X,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[6,-9]`
7	$[-X,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[13,-23]`
8	$[0,\ -X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[14,-22]`
9	$[X,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[15,-24]`
10	$[X,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[16,-20]`
11	$[0,\ X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[17,-19]`
12	$[-X,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[18,-21]`

Table 13: Wyckoff bond: `12c@4c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ 2X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,11]`
2	$[-2X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[2,10]`
3	$[X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[3,12]`
4	$[-X,\ -2X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[4,8]`
5	$[2X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[5,7]`
6	$[-X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[6,9]`
7	$[-X,\ -2X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[13,23]`
8	$[2X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[14,22]`
9	$[-X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[15,24]`
10	$[X,\ 2X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[16,20]`
11	$[-2X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[17,19]`
12	$[X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[18,21]`

Table 14: Wyckoff bond: `24d@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[13]`
14	$[Y, -X+Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[14]`
15	$[X-Y, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[15]`
16	$[X, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[16]`
17	$[-Y, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[17]`
18	$[-X+Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[18]`
19	$[-Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[19]`
20	$[-X+Y, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[20]`
21	$[X, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[21]`
22	$[Y, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[22]`
23	$[X-Y, -Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[23]`
24	$[-X, -X+Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[24]`

* Wyckoff site: `4d`, site symmetry: `-6..`

Table 15: Wyckoff bond: `4a@4d`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,2,3,-16,-17,-18]`
2	$[0, 0, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,5,6,-13,-14,-15]`
3	$[0, 0, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[7,8,9,-22,-23,-24]`
4	$[0, 0, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[10,11,12,-19,-20,-21]`

Table 16: Wyckoff bond: `12b@4d`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,16]`
2	$[-Y, X-Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[2,17]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[3,18]`
4	$[-X, -Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,13]`
5	$[Y, -X+Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[5,14]`
6	$[X-Y, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[6,15]`
7	$[Y, X, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[7,22]`

continued ...

Table 16

No.	vector	center	mapping
8	$[X-Y, -Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[8,23]
9	$[-X, -X+Y, 0]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[9,24]
10	$[-Y, -X, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[10,19]
11	$[-X+Y, Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[11,20]
12	$[X, X-Y, 0]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[12,21]

Table 17: Wyckoff bond: 24c@4d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1]
2	$[-Y, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[2]
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[3]
4	$[-X, -Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[4]
5	$[Y, -X+Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[5]
6	$[X-Y, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[6]
7	$[Y, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[7]
8	$[X-Y, -Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[8]
9	$[-X, -X+Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[9]
10	$[-Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[10]
11	$[-X+Y, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[11]
12	$[X, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[12]
13	$[-X, -Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[13]
14	$[Y, -X+Y, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[14]
15	$[X-Y, X, -Z]$	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[15]
16	$[X, Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[16]
17	$[-Y, X-Y, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[17]
18	$[-X+Y, -X, -Z]$	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[18]
19	$[-Y, -X, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[19]
20	$[-X+Y, Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[20]
21	$[X, X-Y, Z]$	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[21]
22	$[Y, X, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[22]
23	$[X-Y, -Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[23]
24	$[-X, -X+Y, Z]$	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[24]

* Wyckoff site: 4e, site symmetry: 6..

Table 18: Wyckoff bond: 4a@4e, bond symmetry: 6..

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	[1,2,3,4,5,6]
2	$[0, 0, -Z]$	$\left[0, 0, \frac{1}{2}-z\right]$	[7,8,9,10,11,12]
3	$[0, 0, -Z]$	$[0, 0, -z]$	[13,14,15,16,17,18]
4	$[0, 0, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	[19,20,21,22,23,24]

Table 19: Wyckoff bond: `12b@4e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, 0, z]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[0, 0, z]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{2}-z]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$[0, 0, \frac{1}{2}-z]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$[0, 0, \frac{1}{2}-z]$	`[9,-12]`
7	$[-X, -Y, 0]$	$[0, 0, -z]$	`[13,-16]`
8	$[Y, -X+Y, 0]$	$[0, 0, -z]$	`[14,-17]`
9	$[X-Y, X, 0]$	$[0, 0, -z]$	`[15,-18]`
10	$[-Y, -X, 0]$	$[0, 0, z+\frac{1}{2}]$	`[19,-22]`
11	$[-X+Y, Y, 0]$	$[0, 0, z+\frac{1}{2}]$	`[20,-23]`
12	$[X, X-Y, 0]$	$[0, 0, z+\frac{1}{2}]$	`[21,-24]`

Table 20: Wyckoff bond: `24c@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[0, 0, z]$	`[3]`
4	$[-X, -Y, Z]$	$[0, 0, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, 0, z]$	`[5]`
6	$[X-Y, X, Z]$	$[0, 0, z]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{2}-z]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[0, 0, \frac{1}{2}-z]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[0, 0, \frac{1}{2}-z]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{2}-z]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[0, 0, \frac{1}{2}-z]$	`[11]`
12	$[X, X-Y, -Z]$	$[0, 0, \frac{1}{2}-z]$	`[12]`
13	$[-X, -Y, -Z]$	$[0, 0, -z]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[0, 0, -z]$	`[14]`
15	$[X-Y, X, -Z]$	$[0, 0, -z]$	`[15]`
16	$[X, Y, -Z]$	$[0, 0, -z]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[0, 0, -z]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[0, 0, -z]$	`[18]`
19	$[-Y, -X, Z]$	$[0, 0, z+\frac{1}{2}]$	`[19]`
20	$[-X+Y, Y, Z]$	$[0, 0, z+\frac{1}{2}]$	`[20]`
21	$[X, X-Y, Z]$	$[0, 0, z+\frac{1}{2}]$	`[21]`
22	$[Y, X, Z]$	$[0, 0, z+\frac{1}{2}]$	`[22]`
23	$[X-Y, -Y, Z]$	$[0, 0, z+\frac{1}{2}]$	`[23]`
24	$[-X, -X+Y, Z]$	$[0, 0, z+\frac{1}{2}]$	`[24]`

* Wyckoff site: `6f`, site symmetry: `222`

Table 21: Wyckoff bond: `6a@6f`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[1,-4,-8,11]`
2	$[-2X, -X, 0]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2,-5,-7,10]`
3	$[X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,-6,-9,12]`
4	$[-X, -2X, 0]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[13,-16,-20,23]`
5	$[2X, X, 0]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[14,-17,-19,22]`
6	$[-X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[15,-18,-21,24]`

Table 22: Wyckoff bond: `6b@6f`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[1,-4,8,-11]`
2	$[0, X, 0]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2,-5,7,-10]`
3	$[-X, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,-6,9,-12]`
4	$[-X, 0, 0]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[13,-16,20,-23]`
5	$[0, -X, 0]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[14,-17,19,-22]`
6	$[X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[15,-18,21,-24]`

Table 23: Wyckoff bond: `6c@6f`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[1,4,-8,-11]`
2	$[0, 0, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2,5,-7,-10]`
3	$[0, 0, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,6,-9,-12]`
4	$[0, 0, -Z]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[13,16,-20,-23]`
5	$[0, 0, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[14,17,-19,-22]`
6	$[0, 0, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[15,18,-21,-24]`

Table 24: Wyckoff bond: `12d@6f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[1,-4]`
2	$[-Y, X-Y, 0]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2,-5]`
3	$[-X+Y, -X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,-6]`
4	$[Y, X, 0]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[7,-10]`
5	$[X-Y, -Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[8,-11]`
6	$[-X, -X+Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[9,-12]`
7	$[-X, -Y, 0]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[13,-16]`
8	$[Y, -X+Y, 0]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[14,-17]`
9	$[X-Y, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[15,-18]`
10	$[-Y, -X, 0]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[19,-22]`
11	$[-X+Y, Y, 0]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[20,-23]`
12	$[X, X-Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[21,-24]`

Table 25: Wyckoff bond: `12e@6f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 2X, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[1,-8]`
2	$[-2X, -X, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2,-7]`
3	$[X, -X, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,-9]`
4	$[-X, -2X, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[4,-11]`
5	$[2X, X, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[5,-10]`
6	$[-X, X, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[6,-12]`
7	$[-X, -2X, -Z]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[13,-20]`
8	$[2X, X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[14,-19]`
9	$[-X, X, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[15,-21]`
10	$[X, 2X, -Z]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[16,-23]`
11	$[-2X, -X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[17,-22]`
12	$[X, -X, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[18,-24]`

Table 26: Wyckoff bond: `12f@6f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[1,-11]`
2	$[0, X, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2,-10]`
3	$[-X, -X, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3,-12]`
4	$[-X, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[4,-8]`
5	$[0, -X, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[5,-7]`
6	$[X, X, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[6,-9]`
7	$[-X, 0, -Z]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[13,-23]`
8	$[0, -X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[14,-22]`
9	$[X, X, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[15,-24]`
10	$[X, 0, -Z]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[16,-20]`
11	$[0, X, -Z]$	$[0, \frac{1}{2}, \frac{3}{4}]$	`[17,-19]`
12	$[-X, -X, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[18,-21]`

Table 27: Wyckoff bond: `24g@6f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[1]`
2	$[-Y, X-Y, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2]`
3	$[-X+Y, -X, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[3]`
4	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[4]`
5	$[Y, -X+Y, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[5]`
6	$[X-Y, X, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[6]`
7	$[Y, X, -Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[7]`
8	$[X-Y, -Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[8]`
9	$[-X, -X+Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[9]`
10	$[-Y, -X, -Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[10]`
11	$[-X+Y, Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[11]`
12	$[X, X-Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]$	`[12]`

continued ...

Table 27

No.	vector	center	mapping
13	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[13]`
14	$[Y, -X+Y, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[14]`
15	$[X-Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15]`
16	$[X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[16]`
17	$[-Y, X-Y, -Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[17]`
18	$[-X+Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[18]`
19	$[-Y, -X, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[19]`
20	$[-X+Y, Y, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[20]`
21	$[X, X-Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[21]`
22	$[Y, X, Z]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[22]`
23	$[X-Y, -Y, Z]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[23]`
24	$[-X, -X+Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[24]`

* Wyckoff site: `6g`, site symmetry: `2/m..`

Table 28: Wyckoff bond: `6a@6g`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4,-13,16]`
2	$[-Y, X-Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-5,-14,17]`
3	$[-X+Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6,-15,18]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-10,-19,22]`
5	$[X-Y, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-11,-20,23]`
6	$[-X, -X+Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-12,-21,24]`

Table 29: Wyckoff bond: `6b@6g`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,-13,-16]`
2	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,-14,-17]`
3	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,-15,-18]`
4	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,10,-19,-22]`
5	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,11,-20,-23]`
6	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,12,-21,-24]`

Table 30: Wyckoff bond: `12c@6g`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-13]`
2	$[-Y, X-Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-14]`
3	$[-X+Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-15]`
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-16]`
5	$[Y, -X+Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-17]`
6	$[X-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,-18]`

continued ...

Table 30

No.	vector	center	mapping
7	$[Y,\ X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,-19]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[8,-20]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,-21]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,-22]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[11,-23]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[12,-24]`

* Wyckoff site: `8h`, site symmetry: `3..`

Table 31: Wyckoff bond: `8a@8h`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[4,5,6]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[7,8,9]`
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[10,11,12]`
5	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[13,14,15]`
6	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[16,17,18]`
7	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[19,20,21]`
8	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[22,23,24]`

Table 32: Wyckoff bond: `24b@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[12]`
13	$[-X,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[13]`
14	$[Y,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[14]`
15	$[X-Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[15]`
16	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[16]`
17	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[17]`
18	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[18]`
19	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[19]`
20	$[-X+Y,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[20]`
21	$[X,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[21]`
22	$[Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[22]`

continued ...

Table 32

No.	vector	center	mapping
23	$[X-Y,\,-Y,\,Z]$	$\left[\frac{2}{3},\,\frac{1}{3},\,z+\frac{1}{2}\right]$	`[23]`
24	$[-X,\,-X+Y,\,Z]$	$\left[\frac{2}{3},\,\frac{1}{3},\,z+\frac{1}{2}\right]$	`[24]`

* Wyckoff site: `12i`, site symmetry: `2..`

Table 33: Wyckoff bond: `12a@12i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\,Y,\,0]$	$\left[\frac{1}{2},\,0,\,z\right]$	`[1,-4]`
2	$[-Y,\,X-Y,\,0]$	$\left[0,\,\frac{1}{2},\,z\right]$	`[2,-5]`
3	$[-X+Y,\,-X,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,z\right]$	`[3,-6]`
4	$[Y,\,X,\,0]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}-z\right]$	`[7,-10]`
5	$[X-Y,\,-Y,\,0]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}-z\right]$	`[8,-11]`
6	$[-X,\,-X+Y,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}-z\right]$	`[9,-12]`
7	$[-X,\,-Y,\,0]$	$\left[\frac{1}{2},\,0,\,-z\right]$	`[13,-16]`
8	$[Y,\,-X+Y,\,0]$	$\left[0,\,\frac{1}{2},\,-z\right]$	`[14,-17]`
9	$[X-Y,\,X,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,-z\right]$	`[15,-18]`
10	$[-Y,\,-X,\,0]$	$\left[0,\,\frac{1}{2},\,z+\frac{1}{2}\right]$	`[19,-22]`
11	$[-X+Y,\,Y,\,0]$	$\left[\frac{1}{2},\,0,\,z+\frac{1}{2}\right]$	`[20,-23]`
12	$[X,\,X-Y,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,z+\frac{1}{2}\right]$	`[21,-24]`

Table 34: Wyckoff bond: `12b@12i`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\,0,\,Z]$	$\left[\frac{1}{2},\,0,\,z\right]$	`[1,4]`
2	$[0,\,0,\,Z]$	$\left[0,\,\frac{1}{2},\,z\right]$	`[2,5]`
3	$[0,\,0,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,z\right]$	`[3,6]`
4	$[0,\,0,\,-Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}-z\right]$	`[7,10]`
5	$[0,\,0,\,-Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}-z\right]$	`[8,11]`
6	$[0,\,0,\,-Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}-z\right]$	`[9,12]`
7	$[0,\,0,\,-Z]$	$\left[\frac{1}{2},\,0,\,-z\right]$	`[13,16]`
8	$[0,\,0,\,-Z]$	$\left[0,\,\frac{1}{2},\,-z\right]$	`[14,17]`
9	$[0,\,0,\,-Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,-z\right]$	`[15,18]`
10	$[0,\,0,\,Z]$	$\left[0,\,\frac{1}{2},\,z+\frac{1}{2}\right]$	`[19,22]`
11	$[0,\,0,\,Z]$	$\left[\frac{1}{2},\,0,\,z+\frac{1}{2}\right]$	`[20,23]`
12	$[0,\,0,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,z+\frac{1}{2}\right]$	`[21,24]`

Table 35: Wyckoff bond: `24c@12i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$\left[\frac{1}{2},\,0,\,z\right]$	`[1]`
2	$[-Y,\,X-Y,\,Z]$	$\left[0,\,\frac{1}{2},\,z\right]$	`[2]`
3	$[-X+Y,\,-X,\,Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,z\right]$	`[3]`
4	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{2},\,0,\,z\right]$	`[4]`

continued ...

Table 35

No.	vector	center	mapping
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7]`
8	$[X-Y,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[8]`
9	$[-X,\ -X+Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[9]`
10	$[-Y,\ -X,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10]`
11	$[-X+Y,\ Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[11]`
12	$[X,\ X-Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[12]`
13	$[-X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[13]`
14	$[Y,\ -X+Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[14]`
15	$[X-Y,\ X,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[15]`
16	$[X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[16]`
17	$[-Y,\ X-Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ -z\right]$	`[17]`
18	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[18]`
19	$[-Y,\ -X,\ Z]$	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[19]`
20	$[-X+Y,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[20]`
21	$[X,\ X-Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[21]`
22	$[Y,\ X,\ Z]$	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[22]`
23	$[X-Y,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[23]`
24	$[-X,\ -X+Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[24]`

* Wyckoff site: `12j`, site symmetry: `.2.`

Table 36: Wyckoff bond: `12a@12j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,-8]`
2	$[-2X,\ -X,\ Z]$	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,-7]`
3	$[X,\ -X,\ Z]$	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,-9]`
4	$[-X,\ -2X,\ Z]$	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,-11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[5,-10]`
6	$[-X,\ X,\ Z]$	$\left[x,\ x,\ \frac{1}{4}\right]$	`[6,-12]`
7	$[-X,\ -2X,\ -Z]$	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[13,-20]`
8	$[2X,\ X,\ -Z]$	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[14,-19]`
9	$[-X,\ X,\ -Z]$	$\left[x,\ x,\ \frac{3}{4}\right]$	`[15,-21]`
10	$[X,\ 2X,\ -Z]$	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[16,-23]`
11	$[-2X,\ -X,\ -Z]$	$\left[0,\ x,\ \frac{3}{4}\right]$	`[17,-22]`
12	$[X,\ -X,\ -Z]$	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[18,-24]`

Table 37: Wyckoff bond: `12b@12j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,8]`
2	$[0,\ X,\ 0]$	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,7]`
3	$[-X,\ -X,\ 0]$	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,9]`

continued ...

Table 37

No.	vector	center	mapping
4	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{4}\right]$	`[4,11]`
5	$[0, -X, 0]$	$\left[0, -x, \frac{1}{4}\right]$	`[5,10]`
6	$[X, X, 0]$	$\left[x, x, \frac{1}{4}\right]$	`[6,12]`
7	$[-X, 0, 0]$	$\left[-x, 0, \frac{3}{4}\right]$	`[13,20]`
8	$[0, -X, 0]$	$\left[0, -x, \frac{3}{4}\right]$	`[14,19]`
9	$[X, X, 0]$	$\left[x, x, \frac{3}{4}\right]$	`[15,21]`
10	$[X, 0, 0]$	$\left[x, 0, \frac{3}{4}\right]$	`[16,23]`
11	$[0, X, 0]$	$\left[0, x, \frac{3}{4}\right]$	`[17,22]`
12	$[-X, -X, 0]$	$\left[-x, -x, \frac{3}{4}\right]$	`[18,24]`

Table 38: Wyckoff bond: `24c@12j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[0, x, \frac{1}{4}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[-x, -x, \frac{1}{4}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[x, x, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, x, \frac{1}{4}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[-x, -x, \frac{1}{4}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, x, \frac{1}{4}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[13]`
14	$[Y, -X+Y, -Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[14]`
15	$[X-Y, X, -Z]$	$\left[x, x, \frac{3}{4}\right]$	`[15]`
16	$[X, Y, -Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[16]`
17	$[-Y, X-Y, -Z]$	$\left[0, x, \frac{3}{4}\right]$	`[17]`
18	$[-X+Y, -X, -Z]$	$\left[-x, -x, \frac{3}{4}\right]$	`[18]`
19	$[-Y, -X, Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[19]`
20	$[-X+Y, Y, Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[20]`
21	$[X, X-Y, Z]$	$\left[x, x, \frac{3}{4}\right]$	`[21]`
22	$[Y, X, Z]$	$\left[0, x, \frac{3}{4}\right]$	`[22]`
23	$[X-Y, -Y, Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[23]`
24	$[-X, -X+Y, Z]$	$\left[-x, -x, \frac{3}{4}\right]$	`[24]`

* Wyckoff site: `12k`, site symmetry: `..2`

Table 39: Wyckoff bond: `12a@12k`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[x, 2x, \frac{1}{4}\right]$	`[1,-11]`
2	$[0, X, Z]$	$\left[-2x, -x, \frac{1}{4}\right]$	`[2,-10]`
3	$[-X, -X, Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[3,-12]`

continued ...

Table 39

No.	vector	center	mapping
4	$[-X, 0, Z]$	$\left[-x, -2x, \frac{1}{4}\right]$	`[4,-8]`
5	$[0, -X, Z]$	$\left[2x, x, \frac{1}{4}\right]$	`[5,-7]`
6	$[X, X, Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[6,-9]`
7	$[-X, 0, -Z]$	$\left[-x, -2x, \frac{3}{4}\right]$	`[13,-23]`
8	$[0, -X, -Z]$	$\left[2x, x, \frac{3}{4}\right]$	`[14,-22]`
9	$[X, X, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[15,-24]`
10	$[X, 0, -Z]$	$\left[x, 2x, \frac{3}{4}\right]$	`[16,-20]`
11	$[0, X, -Z]$	$\left[-2x, -x, \frac{3}{4}\right]$	`[17,-19]`
12	$[-X, -X, -Z]$	$\left[x, -x, \frac{3}{4}\right]$	`[18,-21]`

Table 40: Wyckoff bond: `12b@12k`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, 2X, 0]$	$\left[x, 2x, \frac{1}{4}\right]$	`[1,11]`
2	$[-2X, -X, 0]$	$\left[-2x, -x, \frac{1}{4}\right]$	`[2,10]`
3	$[X, -X, 0]$	$\left[x, -x, \frac{1}{4}\right]$	`[3,12]`
4	$[-X, -2X, 0]$	$\left[-x, -2x, \frac{1}{4}\right]$	`[4,8]`
5	$[2X, X, 0]$	$\left[2x, x, \frac{1}{4}\right]$	`[5,7]`
6	$[-X, X, 0]$	$\left[-x, x, \frac{1}{4}\right]$	`[6,9]`
7	$[-X, -2X, 0]$	$\left[-x, -2x, \frac{3}{4}\right]$	`[13,23]`
8	$[2X, X, 0]$	$\left[2x, x, \frac{3}{4}\right]$	`[14,22]`
9	$[-X, X, 0]$	$\left[-x, x, \frac{3}{4}\right]$	`[15,24]`
10	$[X, 2X, 0]$	$\left[x, 2x, \frac{3}{4}\right]$	`[16,20]`
11	$[-2X, -X, 0]$	$\left[-2x, -x, \frac{3}{4}\right]$	`[17,19]`
12	$[X, -X, 0]$	$\left[x, -x, \frac{3}{4}\right]$	`[18,21]`

Table 41: Wyckoff bond: `24c@12k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 2x, \frac{1}{4}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[-2x, -x, \frac{1}{4}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, -2x, \frac{1}{4}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[2x, x, \frac{1}{4}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[2x, x, \frac{1}{4}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[-x, -2x, \frac{1}{4}\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[-2x, -x, \frac{1}{4}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[x, 2x, \frac{1}{4}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$\left[-x, -2x, \frac{3}{4}\right]$	`[13]`
14	$[Y, -X+Y, -Z]$	$\left[2x, x, \frac{3}{4}\right]$	`[14]`
15	$[X-Y, X, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[15]`

continued ...

Table 41

No.	vector	center	mapping
16	$[X, Y, -Z]$	$\left[x, 2x, \frac{3}{4}\right]$	[16]
17	$[-Y, X-Y, -Z]$	$\left[-2x, -x, \frac{3}{4}\right]$	[17]
18	$[-X+Y, -X, -Z]$	$\left[x, -x, \frac{3}{4}\right]$	[18]
19	$[-Y, -X, Z]$	$\left[-2x, -x, \frac{3}{4}\right]$	[19]
20	$[-X+Y, Y, Z]$	$\left[x, 2x, \frac{3}{4}\right]$	[20]
21	$[X, X-Y, Z]$	$\left[x, -x, \frac{3}{4}\right]$	[21]
22	$[Y, X, Z]$	$\left[2x, x, \frac{3}{4}\right]$	[22]
23	$[X-Y, -Y, Z]$	$\left[-x, -2x, \frac{3}{4}\right]$	[23]
24	$[-X, -X+Y, Z]$	$\left[-x, x, \frac{3}{4}\right]$	[24]

* Wyckoff site: 12l, site symmetry: m..

Table 42: Wyckoff bond: 12a@12l, bond symmetry: m..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	[1,16]
2	$[-Y, X-Y, 0]$	$[-y, x-y, 0]$	[2,17]
3	$[-X+Y, -X, 0]$	$[-x+y, -x, 0]$	[3,18]
4	$[-X, -Y, 0]$	$[-x, -y, 0]$	[4,13]
5	$[Y, -X+Y, 0]$	$[y, -x+y, 0]$	[5,14]
6	$[X-Y, X, 0]$	$[x-y, x, 0]$	[6,15]
7	$[Y, X, 0]$	$\left[y, x, \frac{1}{2}\right]$	[7,22]
8	$[X-Y, -Y, 0]$	$\left[x-y, -y, \frac{1}{2}\right]$	[8,23]
9	$[-X, -X+Y, 0]$	$\left[-x, -x+y, \frac{1}{2}\right]$	[9,24]
10	$[-Y, -X, 0]$	$\left[-y, -x, \frac{1}{2}\right]$	[10,19]
11	$[-X+Y, Y, 0]$	$\left[-x+y, y, \frac{1}{2}\right]$	[11,20]
12	$[X, X-Y, 0]$	$\left[x, x-y, \frac{1}{2}\right]$	[12,21]

Table 43: Wyckoff bond: 12b@12l, bond symmetry: m..

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	[1,-16]
2	$[0, 0, Z]$	$[-y, x-y, 0]$	[2,-17]
3	$[0, 0, Z]$	$[-x+y, -x, 0]$	[3,-18]
4	$[0, 0, Z]$	$[-x, -y, 0]$	[4,-13]
5	$[0, 0, Z]$	$[y, -x+y, 0]$	[5,-14]
6	$[0, 0, Z]$	$[x-y, x, 0]$	[6,-15]
7	$[0, 0, -Z]$	$\left[y, x, \frac{1}{2}\right]$	[7,-22]
8	$[0, 0, -Z]$	$\left[x-y, -y, \frac{1}{2}\right]$	[8,-23]
9	$[0, 0, -Z]$	$\left[-x, -x+y, \frac{1}{2}\right]$	[9,-24]
10	$[0, 0, -Z]$	$\left[-y, -x, \frac{1}{2}\right]$	[10,-19]
11	$[0, 0, -Z]$	$\left[-x+y, y, \frac{1}{2}\right]$	[11,-20]
12	$[0, 0, -Z]$	$\left[x, x-y, \frac{1}{2}\right]$	[12,-21]

Table 44: Wyckoff bond: `24c@12l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, 0]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, 0]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -y, 0]$	`[4]`
5	$[Y, -X+Y, Z]$	$[y, -x+y, 0]$	`[5]`
6	$[X-Y, X, Z]$	$[x-y, x, 0]$	`[6]`
7	$[Y, X, -Z]$	$\left[y, x, \frac{1}{2}\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[x-y, -y, \frac{1}{2}\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[-x, -x+y, \frac{1}{2}\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[-x+y, y, \frac{1}{2}\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, x-y, \frac{1}{2}\right]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, -y, 0]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[y, -x+y, 0]$	`[14]`
15	$[X-Y, X, -Z]$	$[x-y, x, 0]$	`[15]`
16	$[X, Y, -Z]$	$[x, y, 0]$	`[16]`
17	$[-Y, X-Y, -Z]$	$[-y, x-y, 0]$	`[17]`
18	$[-X+Y, -X, -Z]$	$[-x+y, -x, 0]$	`[18]`
19	$[-Y, -X, Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[19]`
20	$[-X+Y, Y, Z]$	$\left[-x+y, y, \frac{1}{2}\right]$	`[20]`
21	$[X, X-Y, Z]$	$\left[x, x-y, \frac{1}{2}\right]$	`[21]`
22	$[Y, X, Z]$	$\left[y, x, \frac{1}{2}\right]$	`[22]`
23	$[X-Y, -Y, Z]$	$\left[x-y, -y, \frac{1}{2}\right]$	`[23]`
24	$[-X, -X+Y, Z]$	$\left[-x, -x+y, \frac{1}{2}\right]$	`[24]`

* Wyckoff site: `24m`, site symmetry: `1`

Table 45: Wyckoff bond: `24a@24m`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$[-x, -y, z]$	`[4]`
5	$[Y, -X+Y, Z]$	$[y, -x+y, z]$	`[5]`
6	$[X-Y, X, Z]$	$[x-y, x, z]$	`[6]`
7	$[Y, X, -Z]$	$\left[y, x, \frac{1}{2}-z\right]$	`[7]`
8	$[X-Y, -Y, -Z]$	$\left[x-y, -y, \frac{1}{2}-z\right]$	`[8]`
9	$[-X, -X+Y, -Z]$	$\left[-x, -x+y, \frac{1}{2}-z\right]$	`[9]`
10	$[-Y, -X, -Z]$	$\left[-y, -x, \frac{1}{2}-z\right]$	`[10]`
11	$[-X+Y, Y, -Z]$	$\left[-x+y, y, \frac{1}{2}-z\right]$	`[11]`
12	$[X, X-Y, -Z]$	$\left[x, x-y, \frac{1}{2}-z\right]$	`[12]`
13	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[13]`
14	$[Y, -X+Y, -Z]$	$[y, -x+y, -z]$	`[14]`
15	$[X-Y, X, -Z]$	$[x-y, x, -z]$	`[15]`
16	$[X, Y, -Z]$	$[x, y, -z]$	`[16]`

continued ...

Table 45

No.	vector	center	mapping
17	$[-Y,\, X-Y,\, -Z]$	$[-y,\, x-y,\, -z]$	[17]
18	$[-X+Y,\, -X,\, -Z]$	$[-x+y,\, -x,\, -z]$	[18]
19	$[-Y,\, -X,\, Z]$	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[19]
20	$[-X+Y,\, Y,\, Z]$	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	[20]
21	$[X,\, X-Y,\, Z]$	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	[21]
22	$[Y,\, X,\, Z]$	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[22]
23	$[X-Y,\, -Y,\, Z]$	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$[-X,\, -X+Y,\, Z]$	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[24]